SHUBHAM KUMAR

Phone: 7667956792 | Email: rishab.ss64@gmail.com | Ranchi, Jharkhand | GitHub: github.com/shubham7667

PROFESSIONAL SUMMARY

GATE 2025-qualified M.Tech student in Artificial Intelligence and Machine Learning at BIT Mesra with strong foundations in Machine Learning, NLP, Data Structures, and core Computer Science concepts. Skilled in Python-based data analysis, model development, and problem-solving using modern machine learning libraries.

EDUCATION

M.Tech — AI & ML, BIT Mesra (2025–2027) | CGPA: 8.1
B.Tech — CSE, Sarala Birla University (2021–2025) | CGPA: 7.79
GATE 2025 Qualified

TECHNICAL SKILLS

Programming: Python, C
Libraries: NumPy, Pandas, Scikit-learn, PyTorch (Basics), NLTK
Machine Learning: Regression, Classification, Clustering, Feature Engineering
NLP: TF-IDF, WSD, Text Summarization
Core CS: DSA, OS, OOP
Tools: Git, GitHub, Jupyter Notebook, VS Code, LangChain (Basics)

PROJECTS

Word Sense Disambiguation using Lesk Algorithm

- Developed an NLP system to identify contextual meaning of ambiguous words using Lesk algorithm.
- Applied tokenization, stop-word removal, and context extraction.
- Used lexical overlap for word sense prediction.
- Improved semantic interpretation for NLP tasks.

LULC Analysis using K-Means Clustering

- Built satellite-image-based land classification using Sentinel-2 imagery.
- Analyzed land usage changes across years using geo-coordinates.
- Classified vegetation, water bodies, buildings, barren land.
- Applied false-color visualization for interpretation.

Text Summarization using NLP

- Built extractive summarization pipeline for long documents.
- Used TF-IDF sentence scoring for key sentence selection.
- Reduced document length while preserving important information.
- Improved readability of large textual data.

Car Price Prediction Model

- Built regression model to predict used-car prices.
- Performed preprocessing and feature engineering.
- Applied model evaluation for performance comparison.
- Used EDA for feature analysis.

ACHIEVEMENTS & ACTIVITIES

- Qualified GATE 2025 in CS/IT stream.
- Regularly practice Data Structures and Algorithms.
- Explore Deep Learning, NLP, and LLM applications.